SF Compute Quickstart

ACTION SPACE Hackathon 2026 · train your Isaac Lab policy on a GPU in 8 steps

How it works: you write code on your laptop → upload it to your GPU node → train there → download the trained policy. You never log in to the node: every command is typed on **your laptop** and starts with `autor`.

3 rules: (1) only the `/data` folder on the node is kept, everything else is wiped when it stops. (2) The node costs money every minute it runs: **stop it when you are done**. (3) The node has no screen: train with `--headless`, watch results on your laptop.

STEP 1 Install the tool and log in (once)

```
curl -fsSL https://autoresearch.sfcompute.com/cli.sh | bash
autor login
```

A browser opens: sign in with your team's account. The page then says *Signed in*.

✔ Check: `autor node ls` prints a list (may be empty).

STEP 2 Create your node (once per team)

Replace `myteam` with your team name (lowercase, no spaces).

```
autor node create --name myteam --chip h100 \
  --image nvcr.io/nvidia/isaac-lab:2.3.2 --home /data --wait
```

First time takes a few minutes. **Only use your own node**: never touch another team's.

✔ Check: it ends with `myteam running`.

STEP 3 Every time you open a new terminal: set 2 shortcuts

```
N=myteam
E="--env ACCEPT_EULA=Y --env PRIVACY_CONSENT=Y --env PYTHONUNBUFFERED=1"
```

All commands below use `$N` (your node) and `$E` (Isaac settings).

STEP 4 Check that Isaac Lab works (once)

```
autor run $N -- echo awake
autor run $N $E -- sh -c 'cd /workspace/isaaclab && ./isaaclab.sh -p \
  scripts/reinforcement_learning/rsl_rl/train.py --task Isaac-Cartpole-v0 \
  --headless --max_iterations 50'
```

✔ Check: it prints training blocks up to *Learning iteration 49/50*. Isaac Lab works.

STEP 5 Prepare your project on your laptop

Put everything in one folder, e.g. `myproj/`, with your env/task code and a training script. Two things your training script must do:

• **Make your task known:** add the folder to the Python path, then import your package, e.g. `sys.path.insert(0, "/data/myproj/source/mypkg")` and `import mypkg`.

• **Save logs and checkpoints inside** `/data/myproj/logs` (e.g. a `logs/` folder next to your script). Anywhere else is lost.

Test it on your laptop first with a small run (`--num_envs 16`) if you can.

STEP 6 Upload your project to the node

Run in the folder that **contains** `myproj`:

```
tar --exclude=logs --exclude=.git -czf myproj.tar.gz myproj
autor run $N -- echo awake
autor cp ./myproj.tar.gz $N:/data/myproj.tar.gz
autor run $N -- sh -c 'cd /data && tar -xzf myproj.tar.gz && ls myproj'
```

✔ Check: the last command lists your project files.

Changed one file later? Upload just that file: `autor cp ./myproj/train.py $N:/data/myproj/train.py`

STEP 7 Train

7a. Quick test first (catches errors in 1-2 minutes):

```
autor run $N $E --timeout 600 -- sh -c 'cd /workspace/isaaclab && \
  ./isaaclab.sh -p /data/myproj/train.py --headless --num_envs 64 --max_iterations 5'
```

7b. Real training (runs in the background, `-d`):

```
autor run $N -d $E -- sh -c 'cd /workspace/isaaclab && \
  ./isaaclab.sh -p /data/myproj/train.py --headless --num_envs 4096'
```

7c. Watch it:

```
autor logs $N -f           # live output. Ctrl+C only stops WATCHING, training continues
autor ps $N                # is it still running?  (running / exited)
autor kill $N              # stop the training job
```

If `autor logs` says *couldn't read ... log*: training is usually still fine. Check with `autor ps $N` and look at your checkpoint folder.

STEP 8 Download your trained policy, then STOP the node

```
autor run $N -- sh -c 'ls -R /data/myproj/logs | head -40'     # find your run folder
autor cp $N:/data/myproj/logs/<run-folder> ./myrun              # downloads it (as a tar)
mkdir -p myrun_x && tar -xf myrun -C myrun_x                    # unpack
autor node stop $N                                             # ALWAYS when done
```

Replace `<run-folder>` with the path from the first command. Your trained policy is the `.pt` checkpoint inside (rsl_rl: `model_*.pt`; skrl: `checkpoints/best_agent.pt`). Keep the config yaml files from the run folder too: you need the same inputs and scaling on the drone.

✔ Check: `autor node get $N | grep status` shows *stopped*. Billing is off; your files stay.

If something goes wrong

Problem	Fix
sf login asks to create an organization	Wrong tool. Use `autor login`.
NOTHING WAS WRITTEN ... node is waking	Run `autor run $N -- echo awake`, then repeat.
No module named ... / task not found	Step 5: add the path + import your package in the train script.
My files / logs disappeared	They were not under `/data`.
No output in the logs	You forgot $E (it contains PYTHONUNBUFFERED=1).
Errors about Vulkan / RTX / cameras / video	The H100 cannot render. Use `--headless`, no cameras, no --video.
Command waits forever / times out	Use `-d` (background) + `autor logs $N -f`.
python3: not found (node doctor)	Normal for this image. Always use `./isaaclab.sh -p`.
Still stuck	Ask a host. Bring the exact command and the last 20 lines of output.